How to Build a Universe

Physics, mathematics, logic—and research in the age of AI

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Suppose someone hands you an elegant equation and says: "This could describe the universe."

What would you need before believing them?

You would need more than the equation. You would need to know what can exist, how it changes, what an observer can measure, how probabilities are assigned, and whether the predictions agree with the world. You would also need to know what mathematics was silently assumed along the way.

Was an infinite-dimensional Hilbert space essential, or merely convenient? Was a basis chosen using some form of the Axiom of Choice? Did a finite lattice calculation acquire a continuum interpretation without a proof? Did a Euclidean statistical model quietly become a Lorentzian theory of causes and signals? Did a formal solution become a physical particle?

Our project tries to make those hidden steps visible.

It began as an investigation of ghosts in Weyl gravity. It has grown into a broader programme: **build theories backwards from the jobs physics must do, and record exactly which physical, mathematical, and logical assumptions make each job possible.**

A theory is a stack, not an equation

A useful physical theory needs several layers:

1. **Physical principles.** Symmetry, causality, locality, conservation, and rules for combining systems.
2. **A mathematical world.** Finite sets, Hilbert spaces, Krein spaces, operator algebras, smooth fields, distributions, or internal geometric spaces.
3. **Logical and existence assumptions.** Classical or constructive logic, forms of infinity, compactness, completeness, and choice principles.
4. **Operational bridges.** States, observables, probabilities, preparation, measurement, and detector models.
5. **Contact with evidence.** A prediction with declared inputs and an honest comparison with observations.

Changing any layer can change what follows. Two theories may share the same equation but disagree about states or probability. Two mathematical formulations may describe the same physics—or may only look similar because a missing bridge was assumed.

This is the idea behind **reverse foundations of physics**. Instead of asking only "what does this theory predict?", we also ask:

> What is the weakest, clearest combination of physics, mathematics, and logic that is sufficient for this particular prediction—and which assumptions can be removed by changing representation?

The 576-cell map

The project's interactive atlas crosses three dimensions:

- six mathematical regimes, from ordinary classical mathematics through weakened-choice, constructive, internal, and finite settings;
- six carrier families, including finite algebra, Hilbert space, Krein space, operator algebras, smooth/distributional fields, and localic geometry;
- sixteen physical jobs, including states, dynamics, causality, gauge reduction, interactions, renormalization, reconstruction, and empirical agreement.

That makes $6 \times 6 \times 16 = 576$ coordinates.

They are not 576 rival universes. They are 576 questions. A colored cell means that some scoped evidence has been recorded there. It does not mean the cell is universally solved, and it certainly does not mean neighboring cells can be assembled without proof.

This distinction is surprisingly important. Modern theoretical physics has many beautiful local components. The fragile parts are often the joins:

- from a state to a probability rule;
- from a Euclidean measure to Lorentzian causality;
- from a finite regulator to a continuum theory;
- from gauge cohomology to physical particles;
- from a fitted curve to an explanation;
- from a computer-checked identity to the physical assumptions it was meant to represent.

Several ways to build

The atlas includes familiar and unconventional research traditions.

The mainstream reference

General relativity and quantum field theory provide the calibration baseline. Their broad toolbox has evidence for all sixteen jobs represented in the atlas. But even "16 out of 16" is not a new proof that gravity and quantum theory form one complete theory. The baseline deliberately combines different models and mathematical carriers. Its purpose is to show what mature coverage looks like and to test whether the atlas recognizes known successes.

Two bounded controls reach observations: standard GR passes the declared Cassini light-propagation gate, and GR with an NFW dark-matter halo passes the declared NGC 3198 random-error gate.

Mannheim conformal gravity

Mannheim's programme uses conformal gravity to address galactic rotation without particle dark matter. The linear radial term is genuinely part of the fourth-order vacuum solution. But explaining observations requires more than showing that the term exists. One must decide how matter gets mass, which conformal frame massive bodies follow, what is sourced locally, and what is a global boundary contribution.

On one bounded 39-point NGC 3198 protocol, the Mannheim curve has a slightly smaller ordinary residual than GR+NFW: 4.694 versus 5.148 km/s. Yet the quoted errors change the ordering. GR+NFW has reduced chi-squared 0.965 and passes the declared gate; Mannheim has 3.202 and fails it.

That is not a universal verdict. Distance, inclination, stellar populations, gas profiles, and other systematics are not included. The lesson is more general: "best fit" is meaningless until we say which score, uncertainties, inputs, and scope we mean.

Bateman and Turok's hidden ghost parity

Bateman and Turok propose an unusual route for higher-derivative physics using a positive Euclidean lattice and a hidden indefinite structure. One promising shortcut was to prove that the action gradient always remains at least as strong as the free long-distance scale.

The project found an explicit counterfamily. A carefully compensated source spreads through the periodic Green tail. The fields stay positive, nonseparable, and only polynomially contrasted; their action does not vanish, but the normalized gradient falls below the proposed universal scale.

In ordinary language: **one plausible universal stability shortcut is false.**

This does not refute the full proposal. The dangerous fields might be overwhelmingly improbable, the true interacting operator might control them, or reconstruction might need a different argument. Removing the easy route turns those possibilities into sharper mathematical questions.

Pure-Weyl gravity

Pure Weyl gravity is more symmetric than Einstein gravity and contains four derivatives. Its extra solutions are often called ghosts—but that word hides several different tests:

1. Does an extra local solution exist?
2. Does it survive gauge symmetry, constraints, charges, and boundaries?

3. Can it survive nonlinear backreaction?
4. Can it become a state in a positive quantum theory?

The answers are not all the same. Some extra classical directions survive reduction. Some fail global balance or stability tests. In one Schwarzschild channel, ordinary radiation conditions do not remove the extra layer, and an ordinary black-hole ringing frequency becomes a defective double response pole. That is a sharp mathematical signature, not yet an observable waveform.

The quantum story is equally layered. The strict theory has a nonzero local one-loop anomaly in the setting tested. A compensating field repairs that local identity, but changes the theory. At the causal level, the complete gauge complex admits a BRST-compatible Hadamard two-point **pseudo-state**. It has the right causal singularity and gauge identities, but it is indefinite. It is not a positive physical state, particle theory, or unitarity result.

Constructive, finite, and weak-foundation routes

Finite and constructive models let us ask whether a theorem really needs the usual infinite machinery. An explicitly labelled mode family may avoid a basis-selection step. A finite gauge complex may be checked with exact rational arithmetic. A coded wave can carry computable observables without first constructing the entire smooth continuum.

These are genuine results in their scope. They do not prove that the universe is finite, that all of physics works without Choice, or that a finite model has a continuum limit. "Avoided in this representation" is more honest than "physics disproves the Axiom of Choice."

From ingredients to journeys

To prevent a large matrix from becoming an abstract collection, the atlas also follows eight **theory journeys** through six plain questions:

1. What assumptions does the theory start from?
2. What counts as a state?
3. How does the state change?
4. What can be observed?
5. What numerical prediction is produced?
6. Does it meet a declared empirical benchmark?

Four journeys reach bounded data. GR at Cassini and GR+NFW at NGC 3198 pass their declared gates. Newtonian baryons alone and Mannheim's curve fail the declared NGC 3198 random-error gate. Four other journeys—Bateman–Turok, a free Krein mode, a constructive coded wave, and the pure-Weyl causal quantum route—do not reach an empirical test.

"Not reached" is not "refuted." A later mathematical result also does not repair an earlier missing state, observable, or prediction bridge.

What formal proof adds

Some finite identities are exposed as Lean 4.32/Physlib proof passports. These give reviewers another way to inspect types, signs, and finite algebra. One passport replays a strict-Weyl source identity; another derives a finite graded $q_1/q_2/q_3$ evaluator from operation signatures.

Formalization is valuable because a proof assistant refuses many informal shortcuts. But it proves the statement that was encoded. It cannot decide whether the encoding captures nature, supply a missing physical premise, or turn a finite identity into a continuum quantum theory. The passports are an additional review surface, not a truth machine.

Research in the age of AI

This project is also an experiment in scientific organization. AI systems can derive equations, write software, search literature, challenge claims, and produce fluent papers. They can also produce fluent mistakes, share hidden assumptions, or repeat one another's errors.

The response is not to pretend that AI is an oracle. It is to make the work easier to attack:

```
claim
  → calculation
  → machine-readable certificate
  → independent verifier
  → adversarial test
  → readable paper
  → evidence map
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The repository keeps exact algebra separate from numerical evidence and from exploratory computation. Re-running the same producer is reproduction, not independent verification. Missing inputs, skipped tests, and timeouts are not passes. Negative results and abandoned architectures remain visible because they narrow the search.

Exposition is part of that verification system. If a result cannot be explained clearly enough for another researcher to locate its assumptions, evidence, and failure modes, it is not ready to carry much scientific weight.

How to explore the project

Open the Reverse Physics Atlas and choose the view that matches your question:

- **Dimensions guide:** understand the axes without specialist vocabulary.
- **Matrix:** inspect any of the 576 coordinates.
- **Theory profiles:** compare coverage without confusing it with truth.
- **Assemblies:** see bounded tests, programme prototypes, and missing joins.
- **Theory journeys:** follow ideas toward predictions and observations.
- **Weyl BV routes:** inspect the causal and quantum construction gates.
- **Evidence:** read certificates, literature records, and proof passports.

The technical synthesis is Paper 21. The physicist-level overview is Paper 98. The Bateman–Turok torus paper gives a focused example of the method applied to a proposed universal bound.

What the examples establish

Taken together, the examples establish that several distinctions carry real mathematical content. A classical mode can survive reduction without becoming a positive quantum particle. A causal two-point object can satisfy gauge and Hadamard conditions without defining a positive state. A finite or constructive proof can avoid an existence principle without supplying a continuum limit. A curve can minimize one residual while failing an uncertainty-weighted gate. A universal stability shortcut can fail without the larger research programme being refuted.

The examples also show why assembling a theory is harder than collecting good components. Selected positive structures, nonlinear continuations, changed-theory anomaly repairs, mode-reduced causal responses, and proof-assistant replays each solve a particular job. Unless their domains, carriers, and physical interpretations agree, they do not compose into one universe.

Nothing in the argument supplies a complete theory of nature, a positive full-BV quantum state, Lorentzian interacting quantum gravity, a population-level galactic verdict, a continuum Bateman–Turok reconstruction, or a complete black-hole waveform. Each is a distinct obligation that cannot be inferred from the results presented here.

Bottom line

Building a universe is not choosing one beautiful equation. It is closing a chain of obligations without silently changing the rules halfway through.

The programme's working principle is:

> Separate physical assumptions, mathematical carriers, logical strength, operational bridges, and empirical tests. Then make every connection explicit enough that another person can reproduce it, criticize it, or show exactly where it fails.

That does not guarantee a successful theory. It does turn "interesting idea" into a map of answerable questions.